

GOOGLE PLAY STORE ANALYSIS

App Ecosystem Intelligence

An advanced data-driven exploration of 10,000+ Android applications
analyzing pricing, popularity, and categories with Plotly.

Data Wrangling & Processing

Architecting the ingestion, structural cleaning, and refinement pipelines for reliable analysis.

| Data Ingestion & Quality Cleaning

Structural Purification

We eliminated incomplete data records and dropped non-essential structural parameters (such as 'Last_Updated' and 'Android_Ver'). This streamlined our RAM overhead, allowing fast in-memory execution inside Pandas.

Collision Resolution

Using `.drop_duplicates(subset=['App', 'Type', 'Price'])`, we successfully resolved duplicate records like Instagram, ensuring analytics strictly tracked unique applications without statistical inflation.

Ecosystem Scale Profile

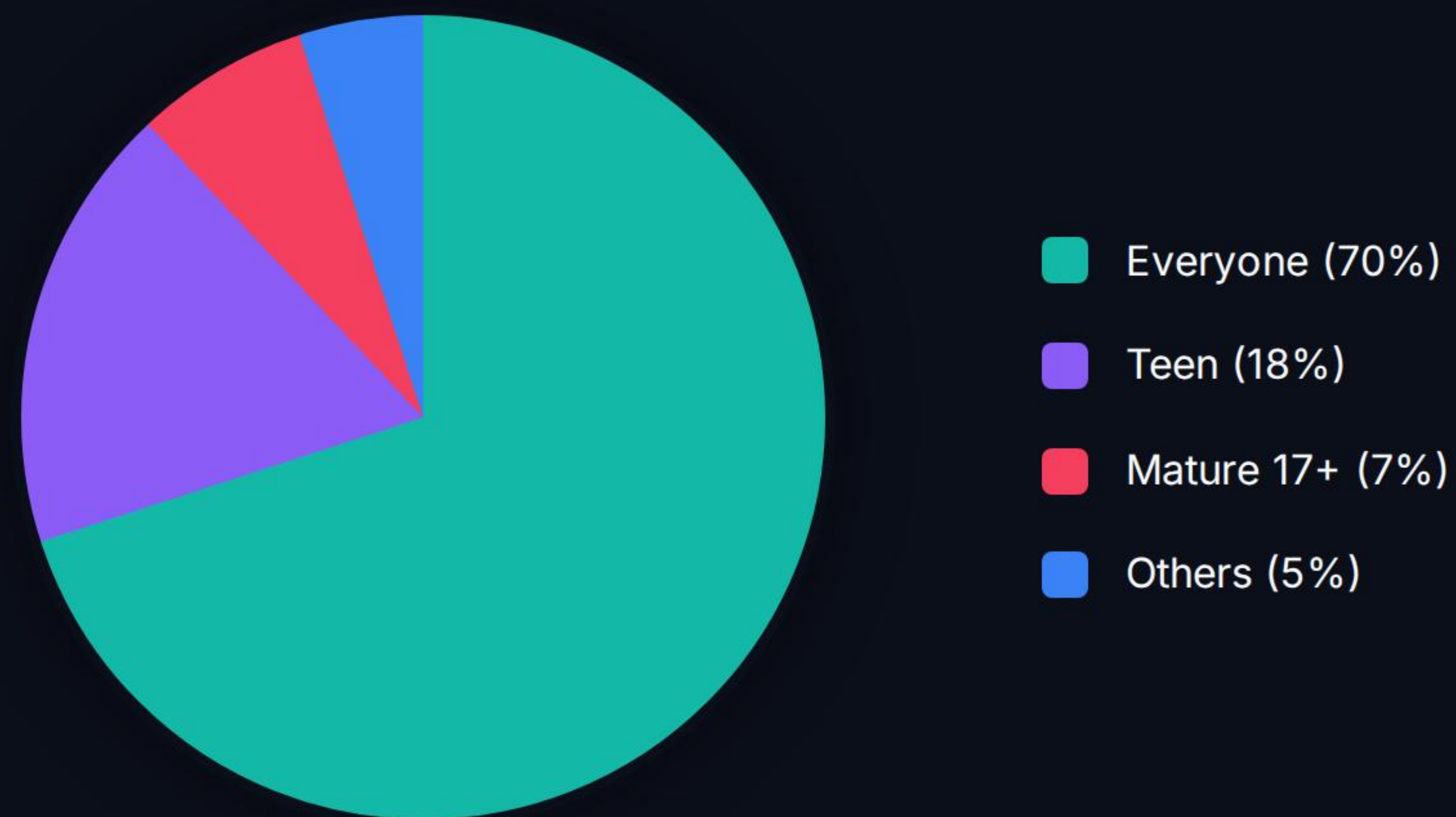
8,199

PROCESSED APPS

Refined Dataset Breadth

Starting with over 10,000 application entries, our cleaning pipeline filtered out corrupt coordinates and duplicated rows. This yielded a perfect structured set of 8,199 unique titles, serving as our reliable operational database.

Content Audience Demographics



The audience distribution is highly centered around general accessibility. Safe, universally rated applications compose 70% of the market space, followed by teen audiences at 18%.

| Advanced Interactive Visuals

The Power of Plotly Express

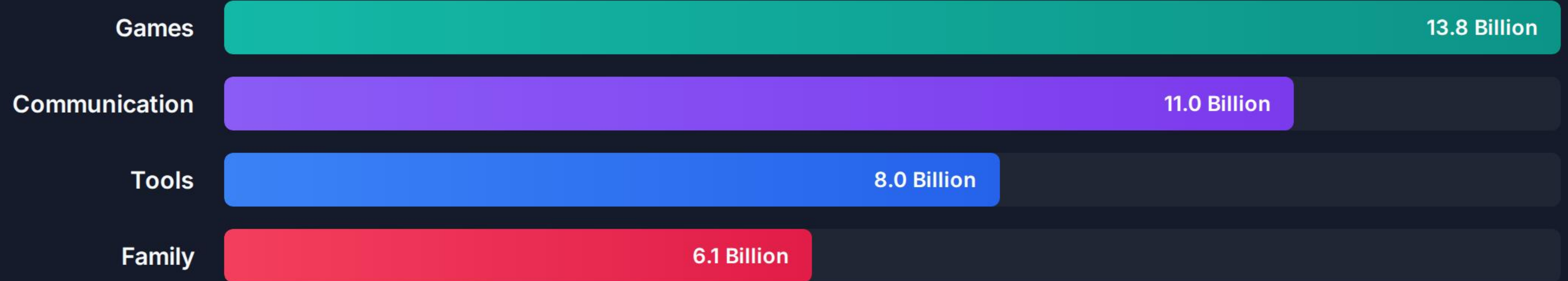
Unlike standard static matplotlib renderings, Plotly shifts the display interface into responsive dimensions. Features like hover labels, dynamic zoom bounds, and high-fidelity output structures allow deeper data investigation.

We utilized Plotly's box, bar, and scatter formats to effortlessly dissect core categories and multi-dimensional pricing thresholds.

Top App Category Concentration

Category	App Count	Market Share %	Dominant Type
Family	1,606	19.59%	Free Model
Game	910	11.10%	In-App Purchases
Tools	719	8.77%	Ad-Supported
Finance	302	3.68%	Secure Transactional

| Install Popularity by Category



Market volume is highly dominated by the Gaming and Communication segments, which secure a combined share of over 24 billion installs across unique records.

| Monetization Structural Insights

- ✔ **Numeric Type Conversion:** We utilized `pd.to_numeric()` after stripping non-numeric formatting characters (such as '\$' and ',') from raw objects to enable clean statistical operations.
- ➔ **Shedding Outliers:** We identified expensive troll apps (like 'I am Rich') priced up to \$400, and filtered our data to `Price < 250` to prevent statistical skewing in averages.
- ★ **Revenue Projections:** Multiplying Installs by Price (`.mul()`) highlighted high-revenue segments, with Medical and Family apps achieving massive monetization.
- 📌 **Free Dominance:** Our comparative analysis confirmed free apps occupy over 92% of the market, representing the dominant strategy for maximizing download volume.

| Paid App Pricing Distribution

Medical



Min: \$0.99 Median: \$5.49 Max: \$79.99

Business



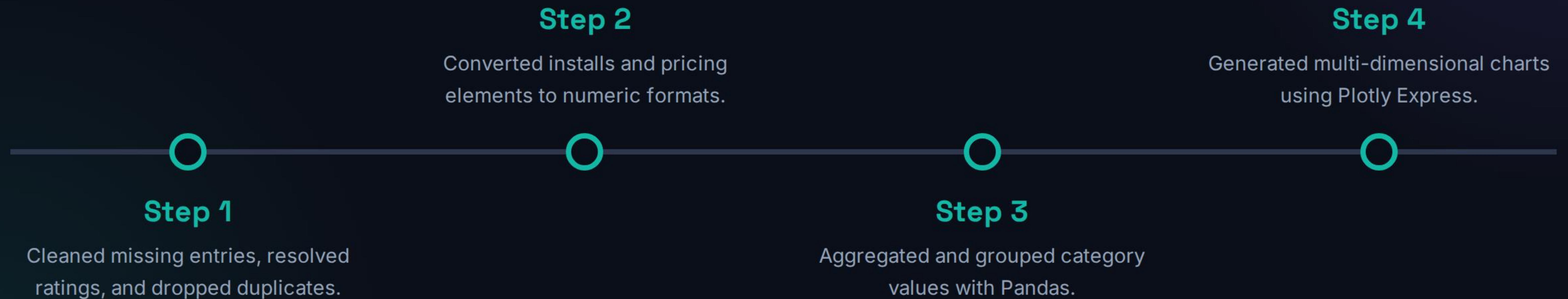
Min: \$0.99 Median: \$4.99 Max: \$29.99

Games



Min: \$0.99 Median: \$2.99 Max: \$14.99

| Analytical Sequence



Questions?

Thank you for exploring this data-driven journey.

QASIM WRITES | DATA INTELLIGENCE CAPSTONE

| Image Sources



https://sleek.design/_next/static/media/nestie-claude-design.161748c61tlrs.webp

Source: sleek.design